

College Management System (SRS)

Software Requirements Specification Document

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Session: 2023-2025

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Revision History

<i>Date</i>	<i>Description</i>	<i>Author</i>	<i>Comments</i>
<date>	<Version 1>	<Sadia Shahbaz>	<First Revision>

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

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Table of Contents

1. Introduction	1
1.1 Purpose	1
1.2 Scope	1
1.3 Definitions, Acronyms, and Abbreviations.	1
1.4 Reference	2
WWW.W3school.com	2
1.5 Overview	2
2. The overall description	3
2.1 Product Perspective	3
2.1.1 Operation:-	3
2.1.2 Site Adaptation Requirement:-	4
2.2 Product Functions	4
2.3 User Characteristics	5
2.4 General Constraints	5
2.5 Assumptions and Dependencies	5
3. Specific Requirements	6
3.1 External Interface Requirements	6
3.1.1 System Interfaces	6
3.1.2 Interfaces	6
3.1.3 Hardware Interfaces	7
3.1.4 Software Interfaces	7
3.1.5 Communications Interfaces	7
3.2 Functional Requirements	7
3.2.1 <Login>	7
3.2.2 <Add student record>	8
3.3 Use Cases	10
3.3.1 Use Case #1	11
3.3.2 Use Case #	12
3.4 Classes / Objects	12
3.4.1 <Class / Object #1>	12
Administration (Class)	12
3.4.1.1 Attributes	12
3.4.1.2 Function	13
3.4.2 <Class / Object #2>	13
Student (Class)	13
3.4.2.1 Attributes	13
3.4.2.2 Functions	13

3.4.3<Class / Object #3>	14
Teacher (Class)	14
3.4.3.1 Attributes	14
3.4.3.2 Functions	14
3.5 Non-Functional Requirements	14
3.5.1 Performance	14
3.5.2 Reliability	14
3.5.3 Availability	15
3.5.4Security	15
3.5.4 Portability	15
3.5.6Maintainability	15
3.6 Inverse Requirements	15
3.7 Logical Database Requirements	15
3.8 Design Constraints	15
3.8.1 Standards Compliance	15
4. Analysis Models	17
4.1 Sequence Diagrams	17
Sequence Diagram	18
4.2 Data Flow Diagrams (DFD)	18
4.2.1 Zero Level Data Flow Diagram	19
4.2.2First Level Data Flow Diagram	20
4.2.3 Second Level Data Flow Diagram	21
4.3 Activity Diagram	21
Activity Diagram	22
4.4 ER Diagram	23
E-R Diagram	24
4.5 Collaboration diagram	24
Components of collaboration diagram:-	24
Collaboration diagram	25
4.6 State-Transition Diagrams (STD)	25
Symbols of state transition diagram:-	26
State transition diagram	26
4.7 Block Diagram	27

CHAPTER 1:

INTRODUCTION

1. Introduction

The title of this project is “College Management System” (CMS). CMS is Desktop application that aims at providing information to all the level of management within an organization. It monitors all aspects of college, its candidates, marks and other activities. It is simple application which allows the user to enter the data and also view its result. CMS is developed to provide the facility to the user to manage the record of the different teachers and students.

1.1 Purpose

The main purpose of developing the “College Management System” (CMS) is to control the task related to the college candidate/employs. It is also used to minimize time to finding to require candidates in college record. “College Management System” (CMS) provides one attractive environment where you can manipulate data and information about candidate and staff easily. In this, the data is stored in an efficient way.

1.2 Scope

“College Management System (CMS)” is used to search information of the user. This is used to login the system and to change the password after logging. This software is used to manage the student attendance and student time table. This software also manages the teacher’s record as well.

The requirement of the user is to:

- Search the information
- Make students attendance
- Make students timetable

1.3 Definitions, Acronyms, and Abbreviations.

Admin: The system who is managing the system software.

User: The person who is using the system.

CMS: College Management System

SRS: Software Requirement Specification.

DBMS: Database Management System

C: C Sharp

GUI: Graphical user interface

SQL: Structured Query language

1.4 Reference

WWW.W3school.com

WWW.google.com

1.5 Overview

The “College Management System (CMS)” can be used to manage the data of all type of educational institute. The main objective of this system is to identify all the functional and N/on functional requirement of the college. It includes information such as external interface Requirements, performance requirement and other technical requirement needed to design the software.

CHAPTER 2:

OVERALL DESCRIPTION

2. The overall description

College Management System(CMS) is a Desktop application. In this, the data will be stored in an organized way i.e. student name, registration no and date of birth. In this system all the data related to school will be stored.

2.1 Product Perspective

- **User Feature:**

The main point for “College Management System” is the college staff, administration and candidate/students. The users for this system are:

- **Administrator:**

Admin: - The super user of the system.

Student: - The users who has the limited access

Staff: - The users who has more rights than a normal users.

2.1.1 Operation:-

The product will cover all the operation on the database:

Database Administrator (DBA) Operation:

Delete none required data manually.

- Update student record.
- Insert new student record.
- Maintain both incoming and outgoing student record.
- Maintain staff record.

Registration Block Operation:-

- See student record.
- Add student record.

- Edit student record.

Student Operation:-

- View only record.

2.1.2 Site Adaptation Requirement:-

In order to run this framework, user must have the PC for working. The requirement for this system is:

- User must have the PC.
- Browser must be introduced on the framework.
- Administrator can update any information.

2.2 Product Functions

In this, the system will be access to specific users like student and staff. Student has to only specific module of the system. The main functions of the “College Management System” are as follows:-

- First we log in.
- Second special ID will be assign to the student, admin and staff.
- When a new student is admitting in college, check student record is already exist OR not. If not exist, make a new account which will be assign a ID and provide all the information like name, father name, date of birth, address and etc.
- When a new lecture is appointed, its registration will be done by DBA.
- Student checks out.
- When student will be passed, its record checked.
- Than the report will be generate. Report will be include all the information about the student like:-
 - Fee report
 - Results
 - Bio data

The administrator has his all the old data related to the college. Every administrator has the basic knowledge of the computer. They are responsible for all the day and night scheduling of staff and lecture.

2.3 User Characteristics

There are two classes of user: admin and user. They should be known about total organization of college management system. They must tight grip on English language. They should also know about computer application. If they should not know how to operate in operating system, they cannot use this software.

2.4 General Constraints

Provide general description of any other items that will limit developer's operations.

There are many constraints of the system such as:

- Graphical User Interface (GUI) should be in English.
- The system should be complete in short time.
- The system is time schedule.
- Admin is responsible for data consistency.
- Delete operation is also available for admin.
- Admin should need login for interaction of system.
- System should not complex.
- User should never be able to break system and to perform any modification.

2.5 Assumptions and Dependencies

There is a list of assumptions and dependencies of this system.

- The user of the system has access to computer and computer must be able to run operating system.
- The system is desktop application so this will run when computer is on.

Chapter 3:

Specification Requirement

3. Specific Requirements

This will be the most important section of the SRS. In this section will give the requirements that are used to guide the projects software design, implementation and testing. Each requirement in this section should be:

- Correct
- Traceable
- Unambiguous
- Verifiable
- Prioritized
- Complete
- Consistent
- Uniquely identifiable

3.1 External Interface Requirements

3.1.1 System Interfaces

In this admin has a login form. Login form contains password and username. It needs a desktop server to run and execute the administration and bring the information from the database to show the clients. It also used to store the different client information.

- User friendly
- Simple
- Attractive
- Self-contained

3.1.2 Interfaces

The Interface is a component of system that interact user. This project basically involves graphical user interface to interact with the user of software system. There are two types of interfaces:

- Hardware interface

- Software interface

3.1.3 Hardware Interfaces

There will be different hardware resources used for this application. That will help to communicate with communication interface.

- Central processing unit
- Processor (1.90GHz)
- RAM(4GB)
- Hard Disk(500MB)
- System Type(64Bit)

3.1.4 Software Interfaces

The software system must be able to communicate with admin in order to manage different things in the system. It must also be able to communicate with the “College Management System” (CMG) in order to manage admission of students. The system must also be able to communicate with the data storage to get the details. There are following software interface:

- Operating system: Window 10
- Language: C sharp
- Front end: Graphical user interface
- Back end: coding in C sharp

3.1.4.1 Microsoft SQL server

Our system is only connecting with SQL.

3.1.5 Communications Interfaces

Our system is interacting with SQL and SQL is interact with database.

3.2 Functional Requirements

Functional requirements of the system are those requirements which describe a function of the system and also explain its components. A function can be defined as a set of different things e.g. input, processing and output. Functional requirement should include.

3.2.1 <Login>

3.2.1.1 Introduction

A “College Management System” can have many functional requirements according to its functionalities and requirements. As there are many functions, there will be as many requirements. In order to meet functional requirements there has to be some inputs. There will be some processing performed on the inputs and after the processing upon the inputs, the output will be provided to the user. There may be some errors in the process, but these errors can be resolved accordingly.

3.2.1.2 Inputs

In order to do different functions of the system, there must be different inputs. For example, if a user wants to register on the system, the user must have to provide his basic information as an input for the registration. If a user wants to login to the system, he must provide to have user name and password as an input.

3.2.1.3 Processing

After providing inputs to the system, processing is done on the inputs in order to gain output. For example, when given the user name and password these inputs are sent to the database and matches with the existing records, then user will have logged in to the system.

3.2.1.4 Outputs

When a user provides input to the system, then the system performs some processing on the inputs. After processing the system will generate the output for the user. For example, when user provides username and password the database processing is done on the input. Then if it matches user will be logged in to the system. Otherwise, the system will generate an error message.

3.2.1.5 Error Handling

If there appears to be any error in the processing of the required function, then it should be handled immediately. There must not be any error while executing of the system.

3.2.2 <Add student record>

3.2.2.1 Introduction

In this step students will be stored in database.

3.2.2.2 Inputs

- Student Name
- Student Roll. No

3.2.2.3 Processing

Check the valid values enter.

3.2.2.4 Output

Message will be shown that record added.

3.2.2.5 Error Handling

Something missing like student name.

3.2.3 <Add Staff Record>

3.2.3.1 Introduction

In this section staff record will be added.

3.2.3.2 Inputs

- Name
- Gender
- Address
- Number
- Email
- Password

3.2.3.3 Processing

System will check these inputs are existing or not.

3.2.3.4 Outputs

Message will be shown staff record updated.

3.2.3.5 Error Handling

Such inputs not exist.

3.2.4<Update>

3.2.4.1 Introduction

The staff or Admin can update in this section.

3.2.4.2 Input

Staff or Admin will enter the student id to update the record.

3.2.4.3 Processing

Student id will compared to the database and return the data of the particular student.

3.2.4.4 Output

The record of the particular student will be displayed.

3.2.4.5 Error handling

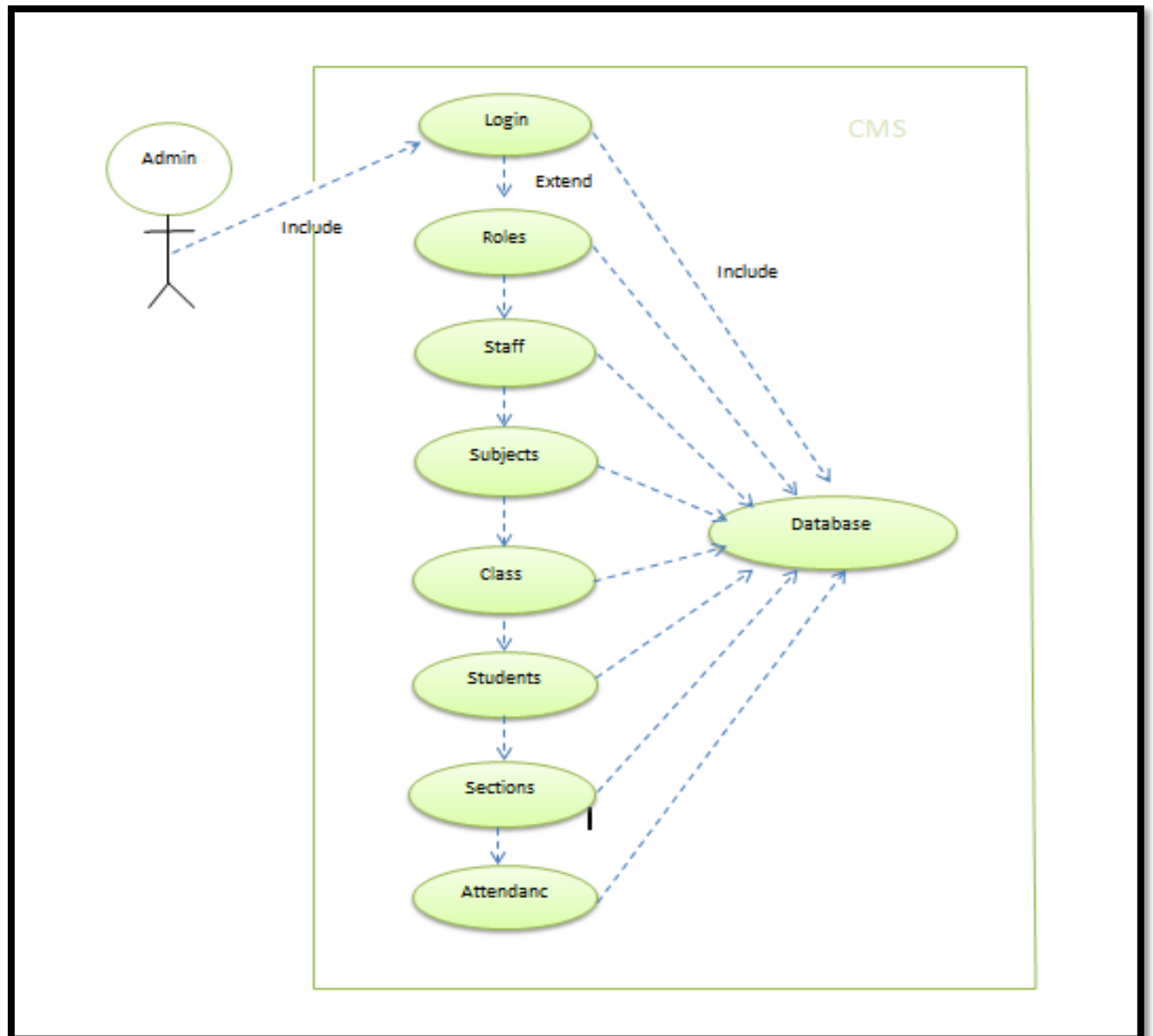
If the student will never be change error message will show.

3.3 Use Cases

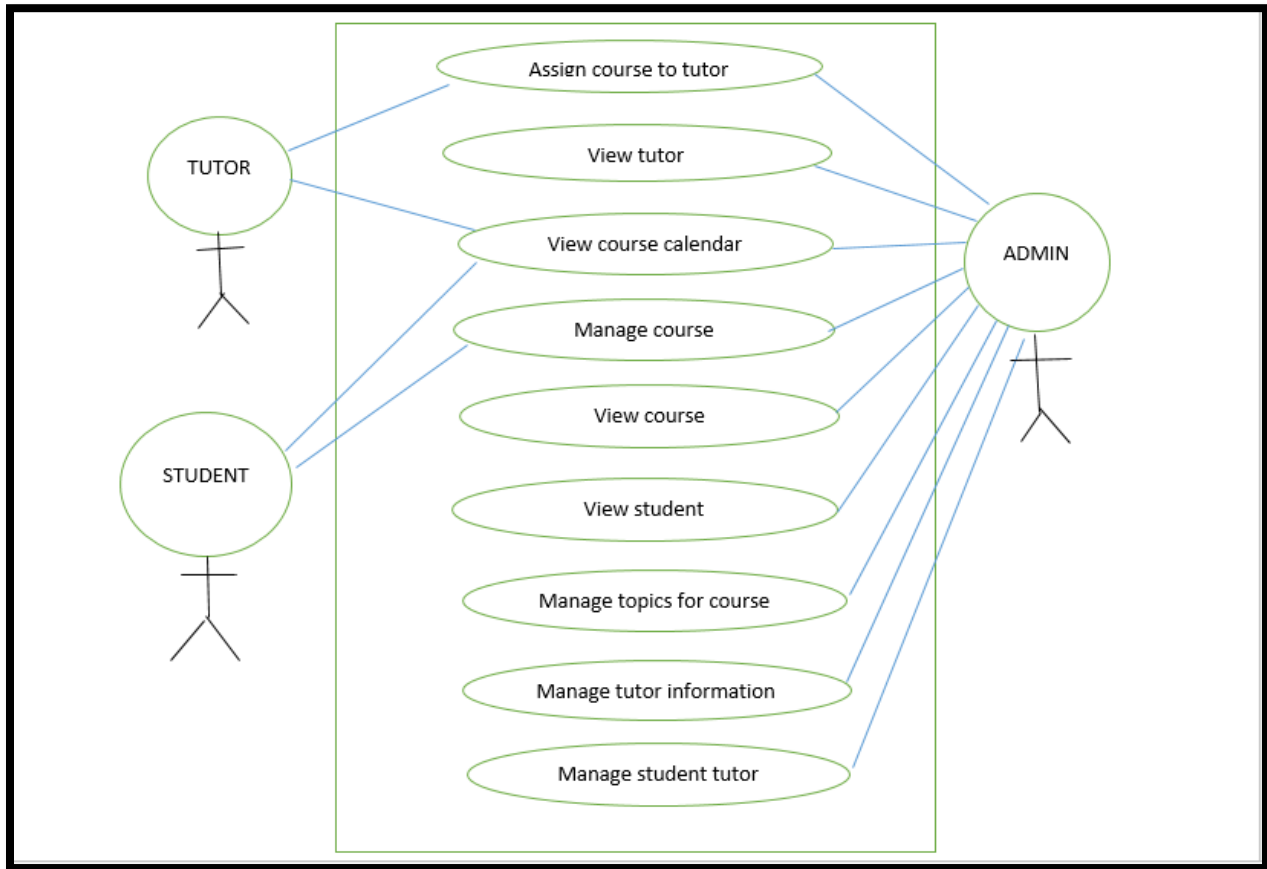
Use cases are basically designed to explain which things will be interacted with the system. It will explain also the different functionalities graphically. Given blow the different terms of use case diagram that will help understand the diagrams to the reader of this document.

Actors	These are the objects that will directly interact with the system. Actors are a person, organization or external system. Actors are shown as stick figures.
Use Cases	These are used to show the functions of the system that are must be implemented. These are oval shapes in the diagram.
Association	Association between actors and use cases are indicated in use case diagram by solid line .
System Boundary	This shows the boundary of the system. It is shown with a large rectangle.
Includes	This shows that a use case uses the functionalities of another use case. This is shown with dotted line and written on it includes.
Extends	This shows that a use case uses all of the other use cases functionality and at adds extra functionality as well. This is also shown with dotted line but written on it extend.

3.3.1 Use Case #1



3.3.2 Use Case



3.4 Classes / Objects

Class defines methods and variables in an object which is a specific entity in a program or unit of a code representing that entity. Class diagrams are useful in all forms of object-oriented.

3.4.1 <Class / Object #1>

Administration (Class)

3.4.1.1 Attributes

- First name
- Last name
- Phone number
- Address

- Postal code
- City
- Scale

3.4.1.2 Function

- Login
- Communication
- Register student or staff
- Add record of particular user
- Delete record of particular user
- Manage record of student as well as teacher

3.4.2 <Class / Object #2>

Student (Class)

3.4.2.1 Attributes

- First name
- Last name
- Phone number
- Address
- Student ID

3.4.2.2 Functions

- Check in
- Check out
- Payment
- Tests
- View their records

3.4.3<Class / Object #3>

Teacher (Class)

3.4.3.1 Attributes

- First name
- Last name
- Phone number
- Address
- City
- Scale
- Staff ID

3.4.3.2 Functions

- Attends the student
- Register student
- Add record to particular student
- Delete record to particular student
- View record to particular student

3.5 Non-Functional Requirements

There are so many requirements for this software system but we will discuss some of them which are most important for College Management System.

3.5.1 Performance

College management system is develop in such a way to that user understand easily.

3.5.2 Reliability

College management system performed function in accurate and compact manner. This system is reliable. Record entry of any student and teacher is accurate and reliable.

3.5.3 Availability

This system is available for user environment. This system is always available to manage the records of student.

3.5.4 Security

The system is secure because unauthorized person cannot use this system. Password is used to login. This system will provide some check system for security.

3.5.4 Portability

This system is portable because it is installed any type of hardware.

3.5.6 Maintainability

This system is maintain regularly and update time to time.

3.6 Inverse Requirements

- CNIC consist of 13 length
- Email must be in this format: Username@gmail.com

3.7 Logical Database Requirements

This section specifies the logical requirements for any information that is to be placed into a database. This may include:

- Types of information used by various functions
- Frequency of use
- Accessing capabilities
- Data entities and their relationships
- Integrity constraints
- Data retention requirements

3.8 Design Constraints

Constraints are followed to develop any system. There are some design constraints to develop any system which are as follows:

3.8.1 Standards Compliance

- All field will be filled correctly.

- User data will be stored in database.
- After completing any operation message will be displayed successful.
- The software is designed in budget, time and resources.

Chapter 4:

Analysis Model

4. Analysis Models

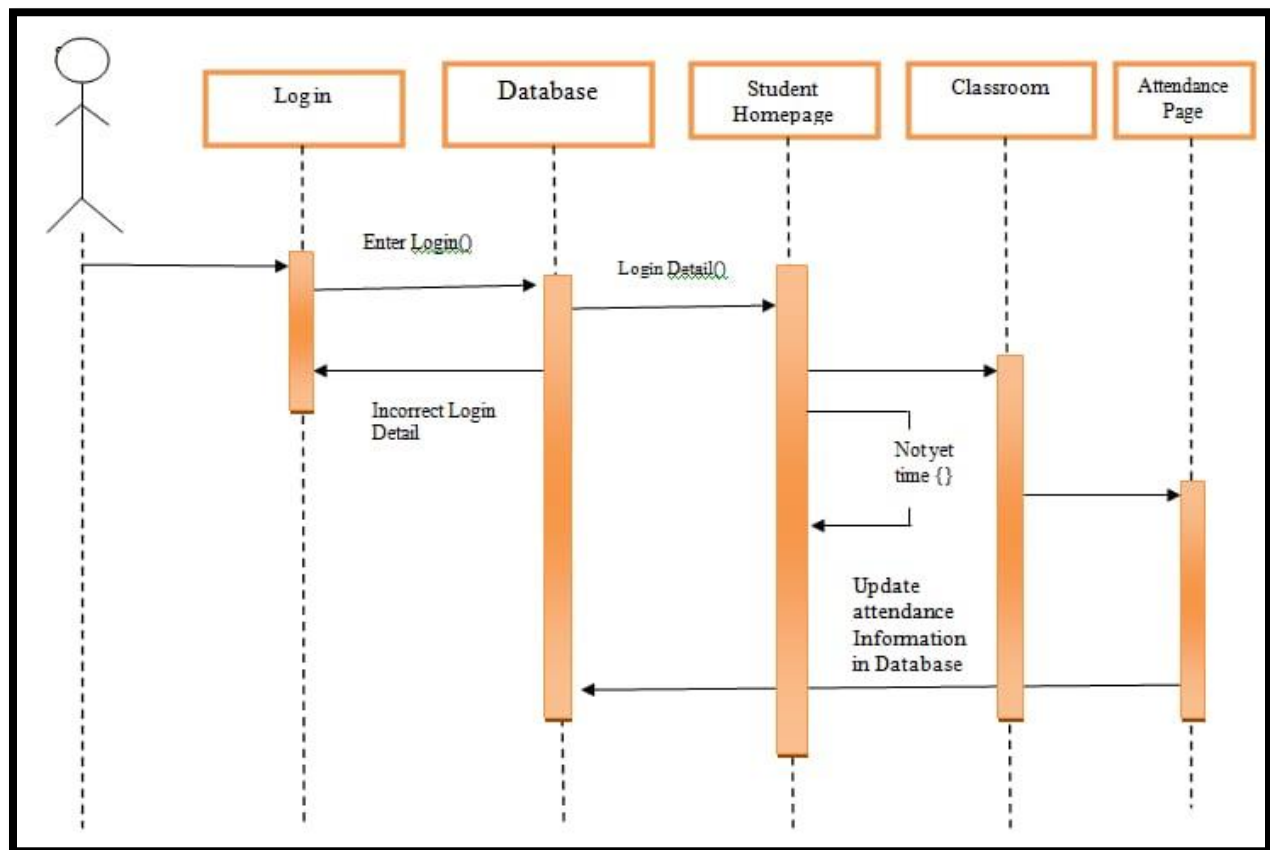
In analysis different diagrams are used for college management system. Each model should include an introduction and narrative description. This section describes the analysis of the project in the real world. Analysis of the project is done by using different diagrams of software engineering.

4.1 Sequence Diagrams

As the name indicates, sequence diagram explain different sequence of different tasks. The purpose of this diagram is to show that the system is running in the proper way and nothing will go wrong at the time of executing.

AXIS	The x-axis shows the movement among different things and the y-axis shows the time.
COMMENT	Comment is used to explain what are going on the specific time or space. These are written on left side of the diagram.
INSTANCE	These are solid boxes on the top of the diagram with dotted lines.
CALL	These are the lines with the triangle at the end of each line. These are used to show movement of one instance to another.
RETURNS	These are used to show the return value of the function. Dotted line with an arrow on the edge is used to show the return.
GUARD STATEMENT	Guard statement is used to set some kind of criteria on any function.

Sequence Diagram



4.2 Data Flow Diagrams (DFD)

A data flow diagram is the graphical representation of flow of data through of information system, modeling its aspects. A DFD shows what kind of information will be input to and output from the system, where the data will come from and go to and where the data will be stored. A DFD is the traditional representation of the information flow within the system. The DFD consist of four major components: entities, process, data store and data flow. The symbols used to depict hoe these components interact in a system are simple and easy.

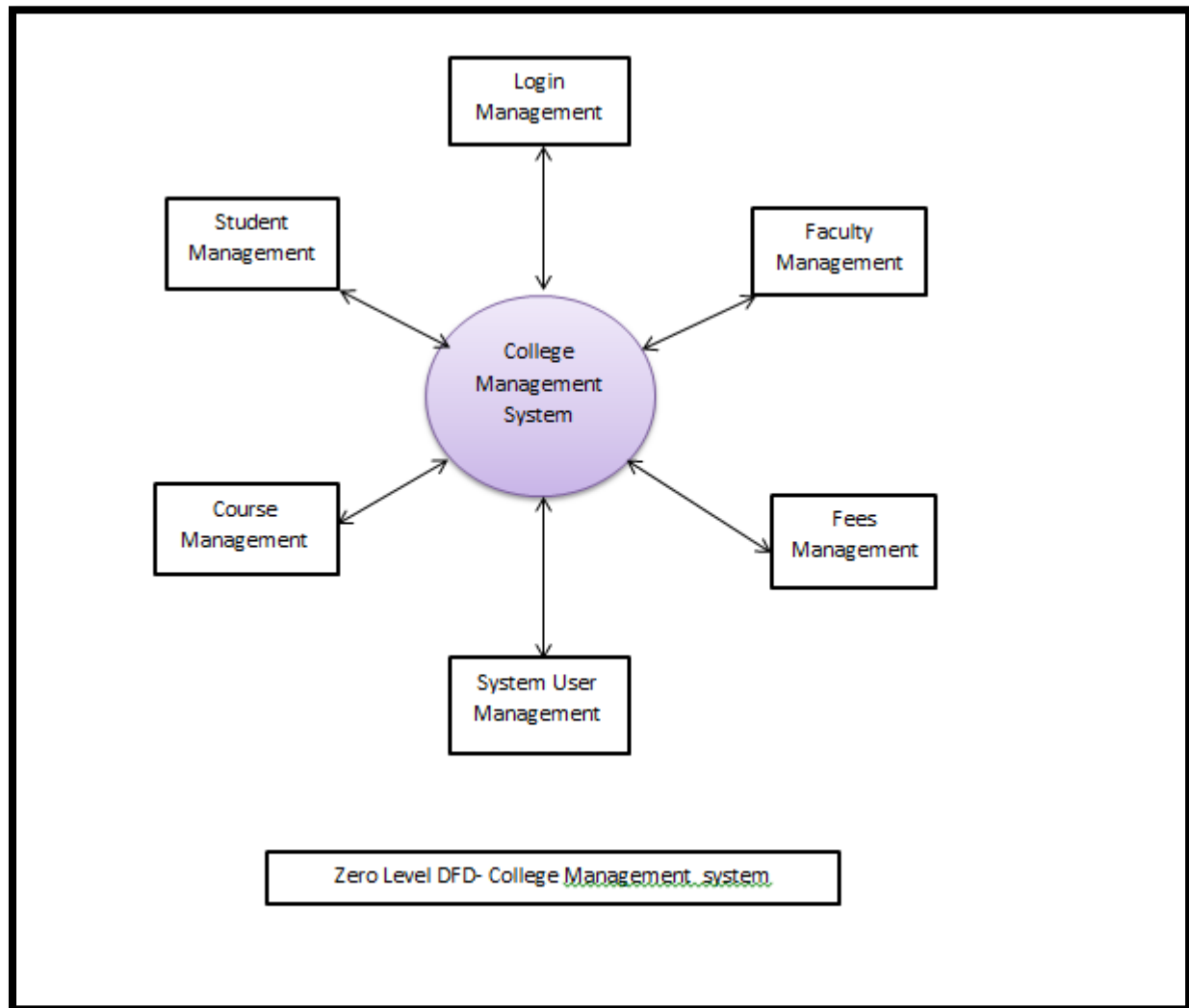
DFD construction:

To draw DFD following components are used:

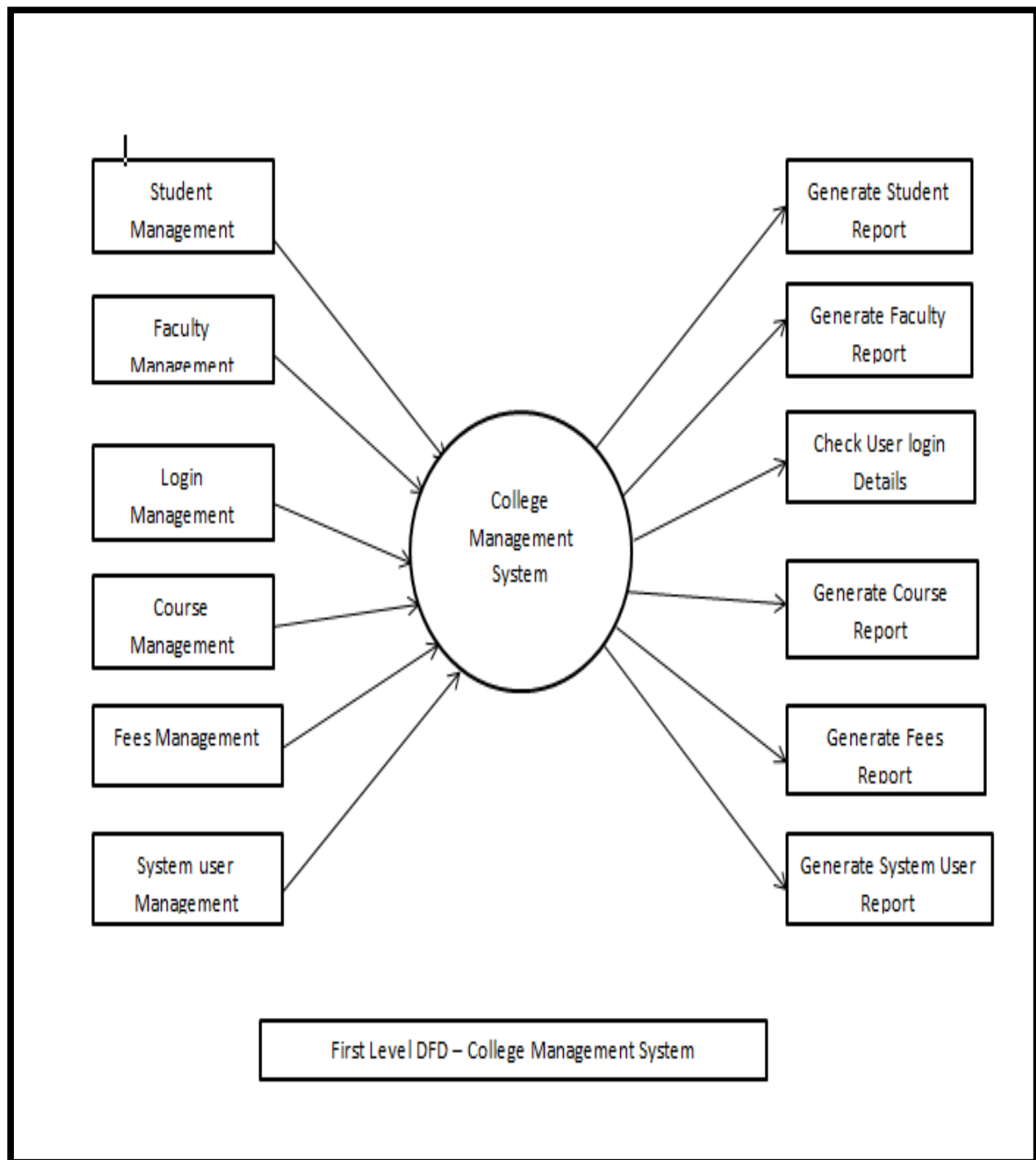
External entity	External entity can be represent as a system or subsystem or human.
Process	Process is the actions that are performed on incoming data to produce some appropriate result.

Data store	Data storage is used to store data permanently or temporary.
Data flow	Data flow lines shows the flow of data that is being processed.

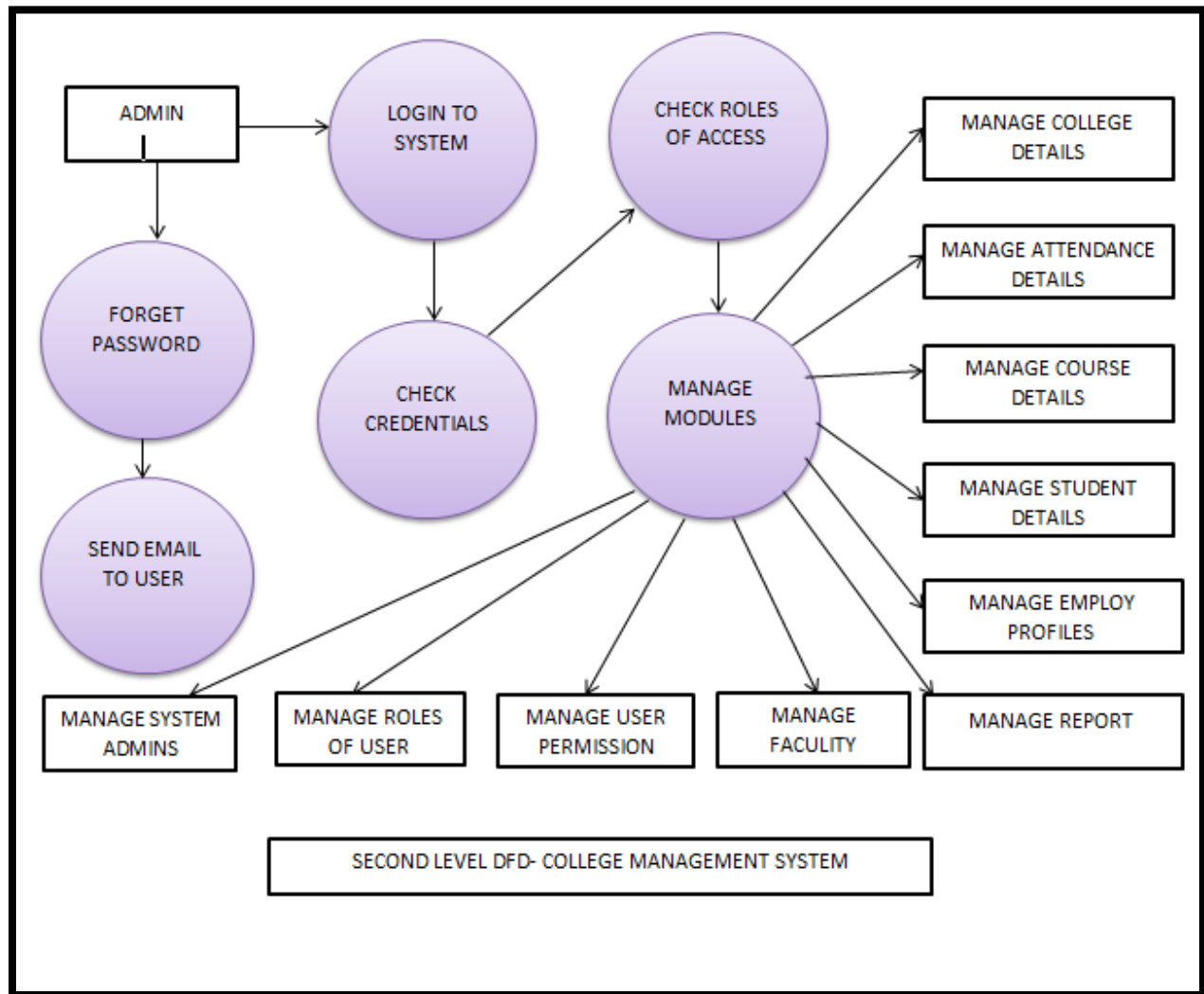
4.2.1 Zero Level Data Flow Diagram



4.2.2 First Level Data Flow Diagram



4.2.3 Second Level Data Flow Diagram



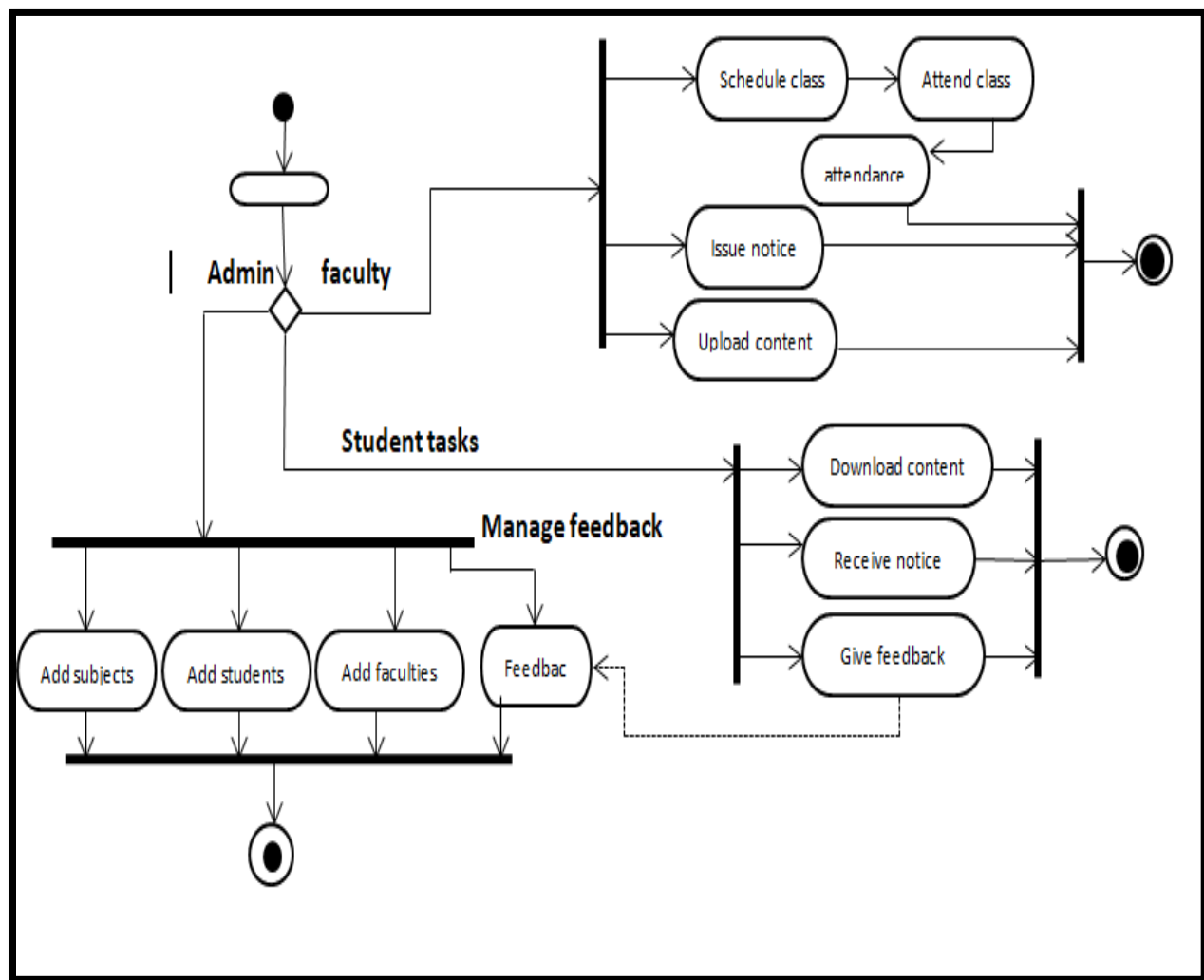
4.3 Activity Diagram

Activity diagrams describe the workflow behavior of the system. These diagrams are used in process modeling and analysis during requirements engineering. They are most useful for understanding work flow analysis of synchronous behaviors across a process. The diagrams describe the state of activities by showing the sequence of activities performed. They can show activities these are parallel or conditional. The main reason to use the activity diagram is to model the workflow behind the system being designed.

Components of Activity diagram:

Initial node	The filled in circle is the starting point of diagram. It does make it easier to read.
Final node	The filled circle with a border is the ending point of diagram. An activity diagram has zero or more ending point.
Flow	The flow is represented with the arrow in the diagram.
Fork	A fork is used when multiple activities are occurring at the same time.
Merge	A diamond with several flows entering and one leaving.
Join	A black bar with several flows and one entering.

Activity Diagram



4.4 ER Diagram

ER stands for “Entity Relationship diagram.” An e-r diagram efficiently shows the relationship between various entities stored in a database. E-R Diagram are used to model real world objects like a person, a car, a company etc. It has some features:-

- E-R diagram are used to represent the E-R model in a database, which makes them easy to convert into relationship.
- E-R diagram provide purpose of real world modeling of objects which makes them intently useful.

E-R diagrams require no technical knowledge & no hardware support.

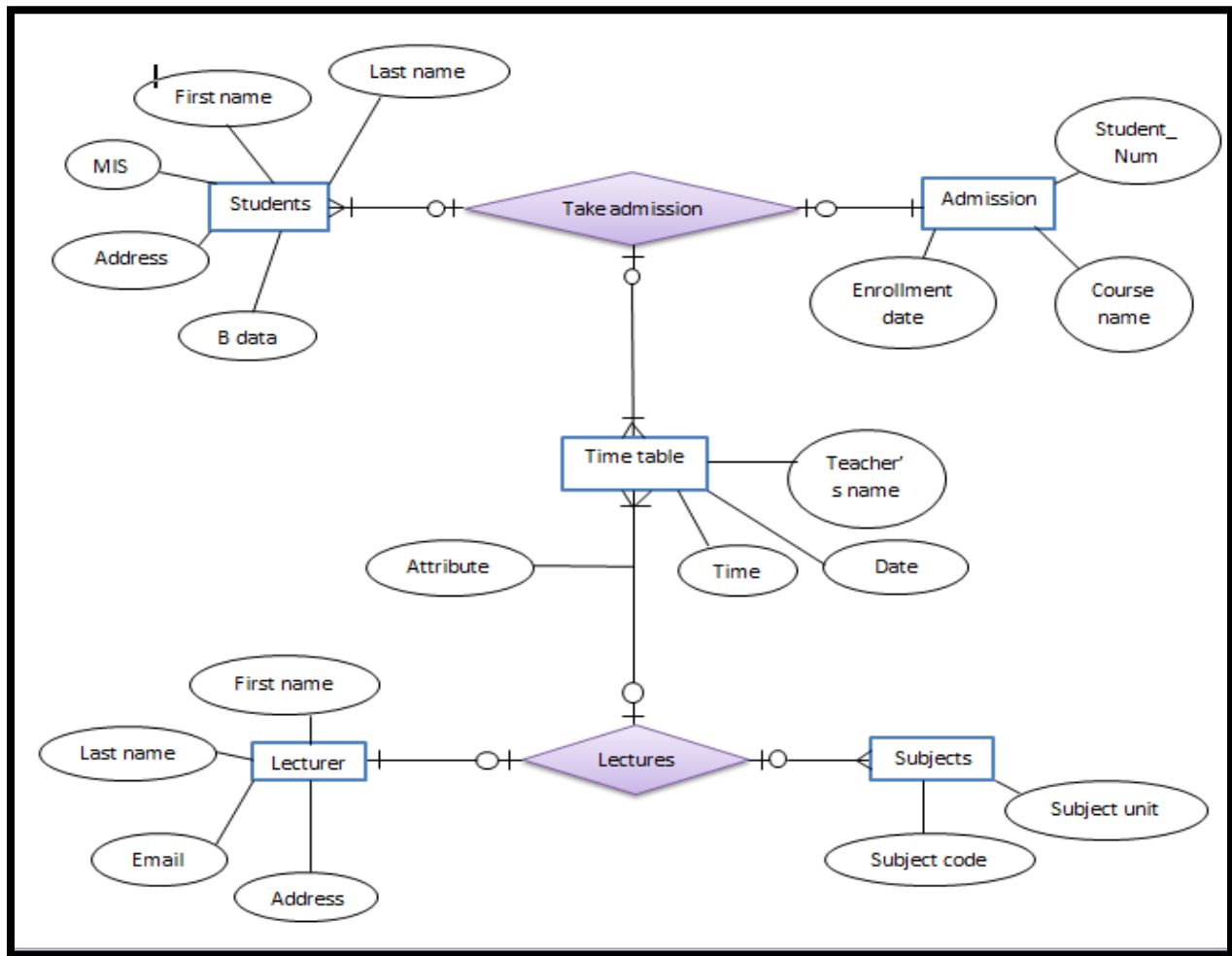
It gives a slandered solution of visualizing the data logically.

These diagrams are very easy to understand.

E-R diagram symbols and annotation:-

	Entity	Any real world object can be represented as an entity which data can be stored in a database.
	Relationship	Any entity has a set of properties. These properties of each entity are termed as attribute.
	Attribute	A relationship is defined as bond or attachment between 2 or more entities.

E-R Diagram



4.5 Collaboration diagram

Collaboration diagram illustrate interaction between objects. It also illustrate message being sent between classes and objects. These are very useful for visualizing the relationship between objects to collaboration to perform particular task. These provide good interaction between objects which may be difficult to see in class level.

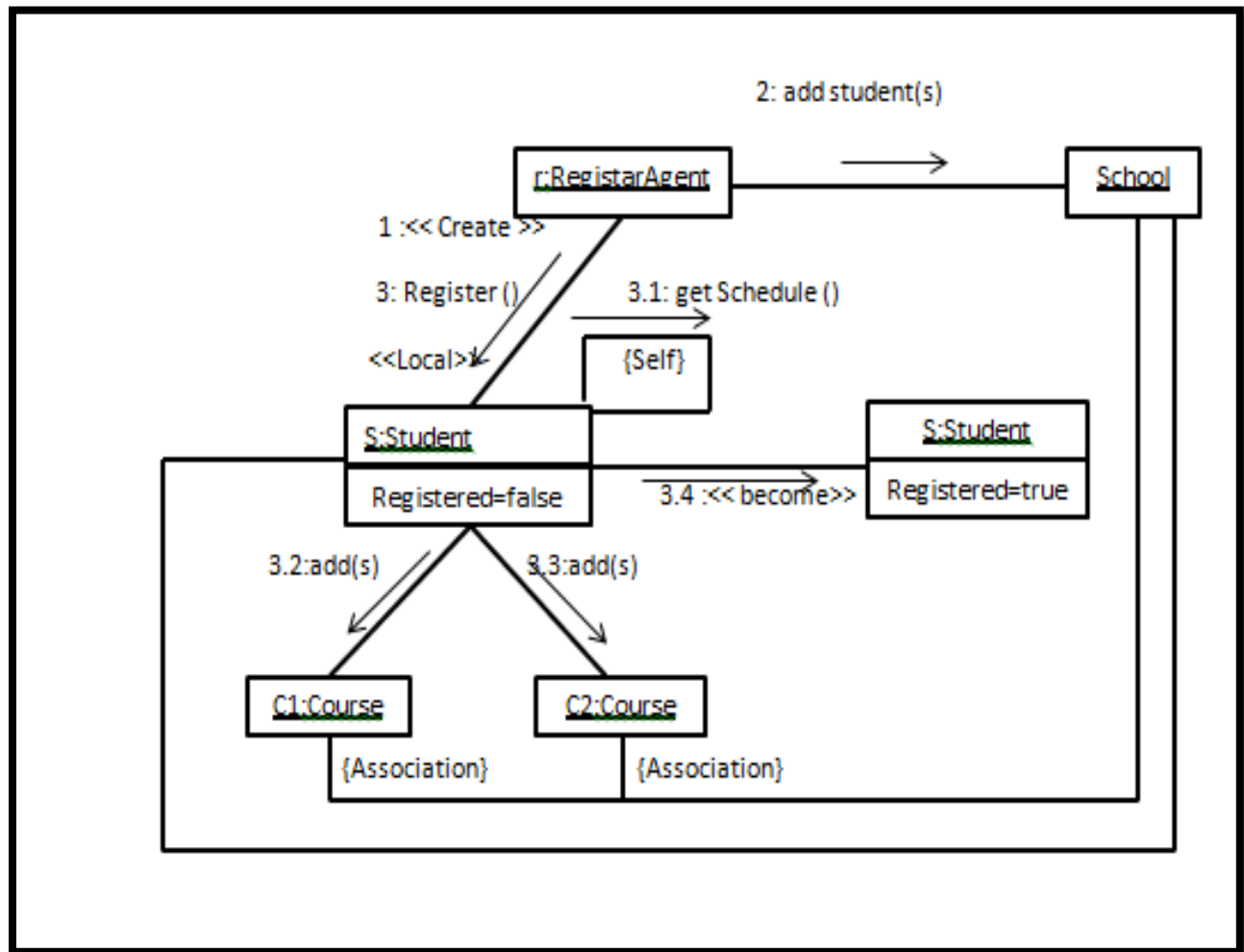
Components of collaboration diagram:-

There are three primary component of collaboration diagram:

- Objects

- Link
- Message

Collaboration diagram



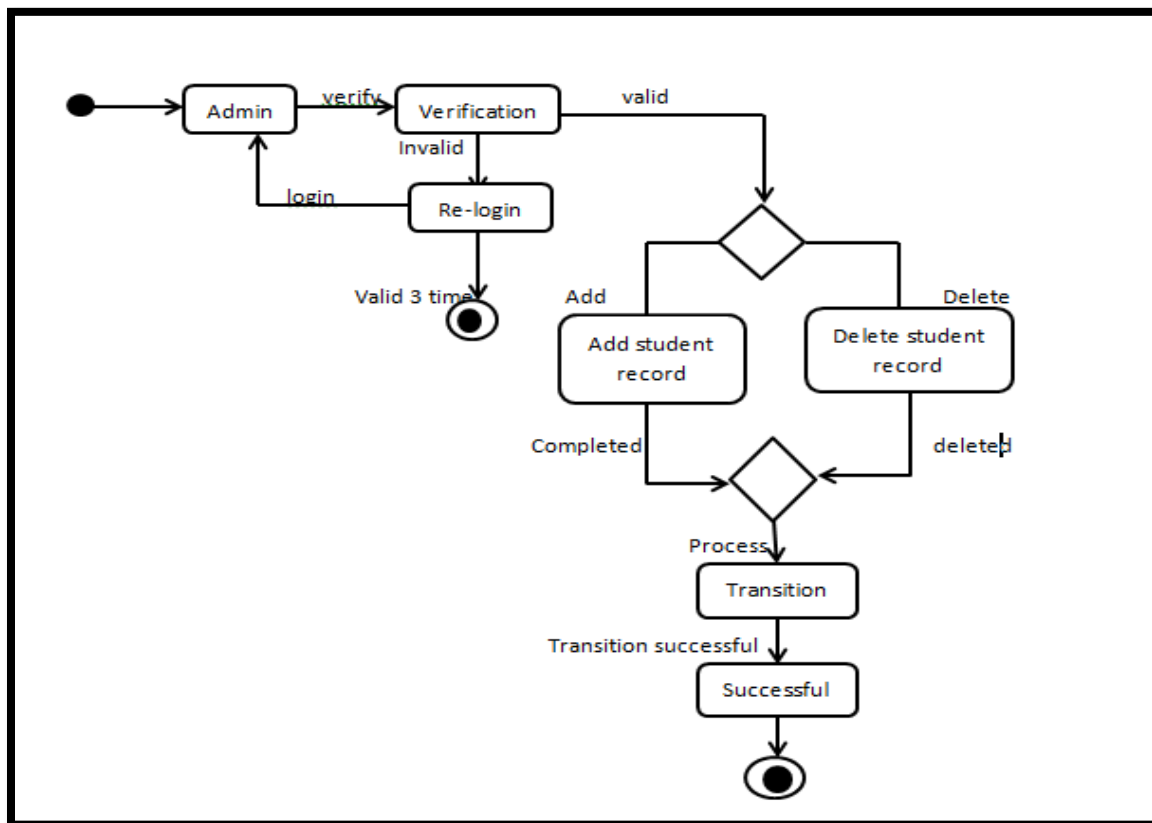
4.6 State-Transition Diagrams (STD)

State transition diagram had a number of states. This receives events from the outside of the world and each event can cause to transition from one state to another. State model describe the behavior of one single object. In state transition diagram we represent that whet actions are leads to change the state of the objects.

Symbols of state transition diagram:-

- State 
- Transition 
- Decision Symbol 
- Start 
- End 

State transition diagram



4.7 Block Diagram

A block diagram is a diagram of a system in which the principal parts are represented by blocks connected by lines that show the relationship of the blocks. They are heavily used in engineering in hardware design, electronic design, and software design and process flow diagrams. The block diagram is used for higher level implementation.

